THE QUIVER ASSOCIATED TO THE STACK OF CERTAIN PERFECT CHAIN COMPLEXES

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ABSTRACT. Let $\mathscr{X}$ be the 2-stack sending a commutative $\mathbb{C}$-algebra to the 2-groupoid of two-term perfect complexes of Euler characteristic zero. We show that the maps $f_n\colon M_n(\mathbb{C}) \to \mathscr{X}$ classifying the tautological complex form a smooth surjective cover, by identifying the fibre products $M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C})$ with a quotient stack $[Z_{n,m}/M_{m,n}]$ of an explicit affine scheme by an additive group. We then describe the category of perverse sheaves on the open substacks $\mathscr{X}_n$.

1. INTRODUCTION

To any dg-category T over a base ring, Toën and Vaquié associate a D^--stack $\mathcal{M}_T$, in the sense of the homotopical algebraic geometry of [TV08], classifying certain T^{op}-dg-modules. When T is saturated, $\mathcal{M}_T$ classifies the compact objects of the triangulated category $[T]$ associated to T, and under finiteness hypotheses on T it is locally geometric: a union of open substacks each of which is geometric [TV07]. The derived algebraic geometry underlying these statements is due to Toën and Vezzosi; [TV03] is a short overview of the framework. We are interested in the representation theory of the geometric substack of [TV07, Example 3.5].

Taking $T = \mathbf{1}$, the stack $\mathcal{M}_{\mathbf{1}}$ classifies perfect complexes up to quasi-isomorphism, and the full substack $\mathcal{M}_{\mathbf{1}}^{[a,b]}$ of those of Tor-amplitude contained in $[a, b]$ is n-geometric for $n = b - a + 1$ [TV07] (see also [HP17, §2.1] for this statement in the form used here). For two-term complexes $[a, b] = [0, 1]$, so $n = 2$: the object we are studying is a 2-stack, not a 1-stack. The same point is visible in the naive picture.

Let $X_n := M_n(\mathbb{C})$ and let $G_n := \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C})$ act on X_n by $(P, Q) \cdot t = QtP^{-1}$, so that two matrices lie in the same orbit exactly when the matrices have the same rank. The space X_n is not literally an open subset of X_{n+1}, and the 1-stack X_n/G_n is not literally an open substack of X_{n+1}/G_{n+1}. To make the "n" version an open subobject of the "$n + 1$" version we pass to 2-stacks; Toën and Vezzosi list the 2-stack of perfect complexes of length 1 among the moduli problems that are not representable even as stacks in groupoids [TV03, §6].

The results are as follows. Section 2 defines the 2-stack $\mathscr{X}$ and shows that the maps $f_n\colon M_n(\mathbb{C}) \to \mathscr{X}$ form a smooth surjective cover, by identifying

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each fibre product $M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C})$ with a quotient stack $[Z_{n,m}/M_{m,n}]$ of an explicit affine scheme by an additive group (Lemma 2.9). Section 3 determines the orbits in $Z_{n,m}$. Section 4 identifies $\mathrm{Perv}(\mathscr{X}_n)$ with $\mathrm{Perv}_{G_n}(M_n(\mathbb{C}))$ and assembles these into a description of $\mathrm{Perv}(\mathscr{X})$.

2. The stack $\mathscr{X}$

Definition 2.1. For a commutative $\mathbb{C}$-algebra A, let $\mathscr{X}(A)$ be the 2-groupoid whose

- objects are the perfect complexes of A-modules of Tor-amplitude contained in $[0, 1]$ and Euler characteristic 0;
- 1-morphisms are the quasi-isomorphisms between such complexes;
- 2-morphisms are the chain homotopies between quasi-isomorphisms.

A ring map $A \to B$ acts by $- \otimes^{\mathbb{L}}_A B$, which makes $\mathscr{X}$ a functor from commutative $\mathbb{C}$-algebras to 2-groupoids. We regard $\mathscr{X}$ as a 2-stack for the fppf topology.

Remark 2.2. Evaluating $\mathscr{X}$ on ordinary commutative $\mathbb{C}$-algebras rather than on cdga's costs nothing here: a derived stack F and its truncation $h^0(F)$ have the same points with values in ordinary commutative algebras [TV03, §3.2]. So $\mathscr{X}$ computes the classical points of $\mathcal{M}_{\mathbf{1}}^{[0,1]}$ cut out by the condition $\chi = 0$.

Let $R_n = \mathbb{C}[x_{ij} \mid 1 \le i, j \le n]$ and consider the map

$$f_n \colon M_n(\mathbb{C}) = \mathrm{Spec}(R_n) \longrightarrow \mathscr{X}$$

corresponding to the chain complex

$$0 \to R_n^{\oplus n} \xrightarrow{\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix}} R_n^{\oplus n} \to 0.$$

For a commutative $\mathbb{C}$-algebra A, the map

$$f_n(A) \colon \mathrm{Spec}(R_n)(A) = \mathrm{Hom}_{\mathbb{C}\text{-alg}}(R_n, A) \longrightarrow \mathscr{X}(A)$$

has essential image the 2-groupoid described as follows.

(1) The objects are the two-term chain complexes

$$\phi \longmapsto 0 \to A^n \xrightarrow{\begin{pmatrix} \phi(x_{11}) & \cdots & \phi(x_{1n}) \\ \vdots & \ddots & \vdots \\ \phi(x_{n1}) & \cdots & \phi(x_{nn}) \end{pmatrix}} A^n \to 0,$$

where $\phi \in \mathrm{Hom}_{\mathbb{C}\text{-alg}}(R_n, A)$ gives A its R_n-module structure.

(2) The 1-morphisms are the quasi-isomorphisms between such complexes.

(3) The 2-morphisms are the chain homotopies between the quasi-isomorphisms.

The essential image of $f_n(A)$ is a full sub-2-groupoid of $\mathscr{X}(A)$: a complex of free modules is cofibrant, so every morphism in $\mathscr{X}(A)$ between two such complexes is represented by an honest chain map, and likewise for 2-morphisms. We will show that the maps $\{f_n\}_{n\in\mathbb{N}}$ give a covering of $\mathscr{X}$.

For any scheme Y and any map g as in the diagram

$$\begin{array}{ccc} M_n(\mathbb{C}) \times_{\mathscr{X}} Y & \xrightarrow{g'} & M_n(\mathbb{C}) \\ \downarrow{\scriptstyle f'_n} & & \downarrow{\scriptstyle f_n} \\ Y & \xrightarrow[g]{} & \mathscr{X} \end{array}$$

our goal is to show that f'_n is smooth and surjective. Since smoothness and surjectivity are local on the target, it suffices to treat the case $Y = \operatorname{Spec}(A)$ for an arbitrary finitely generated $\mathbb{C}$-algebra A.

$$\begin{array}{ccccc} M_n(\mathbb{C}) \times_{\mathscr{X}} Y & \xrightarrow{h'} & M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}) & \xrightarrow{g'} & M_n(\mathbb{C}) \\ \downarrow{\scriptstyle f'_n = h^* f''_n} & & \downarrow{\scriptstyle f''_n} & & \downarrow{\scriptstyle f_n} \\ Y = \operatorname{Spec}(A) & \xrightarrow{h} & M_m(\mathbb{C}) & \xrightarrow{f_m} & \mathscr{X} \end{array}$$

$$Y = \operatorname{Spec}(A) \xrightarrow{g} \mathscr{X}$$

We start by factoring g as $f_m \circ h$ for a suitable m and h; this is Lemma 2.3 below. If f''_n is smooth and surjective, then so is its base change $h^* f''_n = f'_n$.

Lemma 2.3. *Let A be a commutative $\mathbb{C}$-algebra and $g\colon \operatorname{Spec}(A) \to \mathscr{X}$. Then there is a Zariski cover $\operatorname{Spec}(A) = \bigcup_i U_i$ and integers m_i such that $g|_{U_i}$ factors as $f_{m_i} \circ h_i$ for some $h_i\colon U_i \to M_{m_i}(\mathbb{C})$.*

Proof. The map g is the datum of a perfect complex E of A-modules of Tor-amplitude in $[0,1]$ with $\chi(E) = 0$. Such a complex is Zariski-locally on $\operatorname{Spec}(A)$ quasi-isomorphic to a two-term complex $[F^0 \to F^1]$ of finite free modules. Shrinking further so that the ranks are constant, the condition $\chi(E) = 0$ gives $\operatorname{rank} F^0 = \operatorname{rank} F^1 =: m$. A choice of bases identifies the differential with a matrix in $M_m(A)$, i.e. with a map $h\colon \operatorname{Spec}(A) \to M_m(\mathbb{C})$, and by construction the complex it classifies is the one defining g on that open set. Hence $g = f_m \circ h$ there. □

2.1. **Understanding the fibre product.** For a commutative $\mathbb{C}$-algebra A, the fibre product $(M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}))(A)$ is the 1-groupoid described as follows.

(1) The objects are the commutative diagrams

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^n & \xrightarrow{t_0} & A^n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle a_0} & & \downarrow{\scriptstyle a_1} & & \\ 0 & \longrightarrow & A^m & \xrightarrow{t_1} & A^m & \longrightarrow & 0 \end{array}$$

in which (a_0, a_1) is a quasi-isomorphism. We denote such a diagram by the tuple $(t_0, t_1 : a_0, a_1)$.

(2) The morphisms are the chain homotopies between quasi-isomorphisms, i.e.

$$\begin{aligned} &\mathrm{Mor}\big((t_0, t_1 : a_0, a_1),\, (t_0, t_1 : a_0', a_1')\big) \\ &\qquad = \{f \colon A^n \to A^m \mid a_0 - a_0' = f t_0 \text{ and } a_1 - a_1' = t_1 f\}. \end{aligned}$$

This is depicted as

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^n & \xrightarrow{t_0} & A^n & \longrightarrow & 0 \\ & & {\scriptstyle a_0'}\downarrow\downarrow{\scriptstyle a_0} & \swarrow{\scriptstyle f} & {\scriptstyle a_1'}\downarrow\downarrow{\scriptstyle a_1} & & \\ 0 & \longrightarrow & A^m & \xrightarrow[t_1]{} & A^m & \longrightarrow & 0 \end{array}$$

Moreover, if $(t_0, t_1) \neq (t_0', t_1')$ then

$$\mathrm{Mor}\big((t_0, t_1 : a_0, a_1),\, (t_0', t_1' : a_0', a_1')\big) = \emptyset.$$

Lemma 2.4. *Let A be a commutative $\mathbb{C}$-algebra and consider the following diagram, not assumed commutative:*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^n & \xrightarrow{t_0} & A^n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle a_0} & & \downarrow{\scriptstyle a_1} & & \\ 0 & \longrightarrow & A^m & \xrightarrow{t_1} & A^m & \longrightarrow & 0 \end{array}$$

It defines an object of $(M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}))(A)$ if and only if:

(1) $\begin{bmatrix} t_1 & -a_1 \end{bmatrix} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} = t_1 a_0 - a_1 t_0 = 0$*;*

(2) $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$ *is split injective, i.e. it admits a left inverse;*

(3) $\begin{bmatrix} t_1 & -a_1 \end{bmatrix}$ *is surjective.*

Proof. That (a_0, a_1) is a quasi-isomorphism from (A^n, t_0) to (A^m, t_1) is equivalent to the mapping cone being acyclic, i.e. to

$$0 \longrightarrow A^n \xrightarrow{\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}} A^{n+m} \xrightarrow{\begin{bmatrix} t_1 & -a_1 \end{bmatrix}} A^m \longrightarrow 0$$

being a short exact sequence. This gives $\begin{bmatrix} t_1 & -a_1 \end{bmatrix} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} = 0$, injectivity of $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$, surjectivity of $\begin{bmatrix} t_1 & -a_1 \end{bmatrix}$, and $\mathrm{im} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} = \ker \begin{bmatrix} t_1 & -a_1 \end{bmatrix}$.

Since all three terms are free A-modules, the sequence splits, and injectivity of $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$ upgrades to split injectivity. This proves one implication.

Conversely, suppose $\begin{bmatrix} t_1 & -a_1 \end{bmatrix} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} = 0$, that $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$ is split injective and that $\begin{bmatrix} t_1 & -a_1 \end{bmatrix}$ is surjective. As $\begin{bmatrix} t_1 & -a_1 \end{bmatrix}$ is a surjection onto a free module,

$$A^{n+m} \cong \ker \begin{bmatrix} t_1 & -a_1 \end{bmatrix} \oplus A^m.$$

From $\begin{bmatrix} t_1 & -a_1 \end{bmatrix} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} = 0$ we get $\operatorname{im} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} \subseteq \ker \begin{bmatrix} t_1 & -a_1 \end{bmatrix}$, and since $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$ is split injective its image is a direct summand of A^{n+m} contained in the summand $\ker \begin{bmatrix} t_1 & -a_1 \end{bmatrix}$, hence a direct summand of the latter:

$$\ker \begin{bmatrix} t_1 & -a_1 \end{bmatrix} = K \oplus \operatorname{im} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} \cong K \oplus A^n$$

for some A-module K. Combining the two isomorphisms gives $A^{n+m} \cong K \oplus A^n \oplus A^m$. Thus K is a finitely generated projective A-module whose rank at every prime $\mathfrak{p}$ is 0, so $K_{\mathfrak{p}} = 0$ for all $\mathfrak{p}$ and therefore $K = 0$. Hence $\ker \begin{bmatrix} t_1 & -a_1 \end{bmatrix} = \operatorname{im} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$, as required. □

Let $Z_{n,m}$ be the functor of sets given by

$$Z_{n,m}(A) = \big\{\text{objects of } (M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}))(A)\big\}$$

for a commutative $\mathbb{C}$-algebra A. Any set may be regarded as a 1-groupoid by taking its elements as objects, with only identity morphisms. Thus $Z_{n,m}(A)$ is the 1-groupoid with:

(1) objects the tuples $(t_0, t_1 : a_0, a_1)$, i.e. the quasi-isomorphisms (a_0, a_1) between $t_0 \in M_n(A)$ and $t_1 \in M_m(A)$;
(2) morphisms

$$\operatorname{Mor}\big((t_0, t_1 : a_0, a_1),\, (t_0', t_1' : a_0', a_1')\big) = \begin{cases} \{\mathrm{id}\} & \text{if the two tuples agree,} \\ \emptyset & \text{otherwise.} \end{cases}$$

We ask whether $Z_{n,m}$ is representable, and begin with some representable subfunctors. For $I, J \subset \{1, \dots, n+m\}$ with $|I| = n$ and $|J| = m$, set

$$Z_{n,m,I,J} := \operatorname{Spec} \mathbb{C}[t_0, t_1, a_0, a_1, r_1, r_2] \Big/ \Big(\begin{bmatrix} t_1 & -a_1 \end{bmatrix} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}, \; r_1 \det \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}_I - 1, \; r_2 \det \begin{bmatrix} t_1 & -a_1 \end{bmatrix}_J - 1 \Big).$$

Here $\mathbb{C}[t_0, t_1, a_0, a_1, r_1, r_2]$ is the polynomial ring on the variables

$$\begin{aligned} t_0 &= ((t_0)_{ij}), && 1 \le i, j \le n, \\ t_1 &= ((t_1)_{ij}), && 1 \le i, j \le m, \\ a_0 &= ((a_0)_{ij}), && 1 \le i \le m,\ 1 \le j \le n, \\ a_1 &= ((a_1)_{ij}), && 1 \le i \le m,\ 1 \le j \le n, \end{aligned}$$

together with two further variables r_1 and r_2; in particular it is finitely generated.

Lemma 2.5. *The functor $Z_{n,m}$ is representable.*

Proof. For a map $\phi\colon \operatorname{Spec}(R) \to Z_{n,m}$ we have the pullback square

$$\begin{array}{ccc} \phi^{-1}\big(D(\det \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}_I,\ \det \begin{bmatrix} t_1 & -a_1 \end{bmatrix}_J)\big) & \longrightarrow & Z_{n,m,I,J} \\ \downarrow & & \downarrow \\ \operatorname{Spec}(R) & \xrightarrow{\ \phi\ } & Z_{n,m} \end{array}$$

where $D(\det \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}_I,\ \det \begin{bmatrix} t_1 & -a_1 \end{bmatrix}_J)$ is the subfunctor on which $\det \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}_I$ and $\det \begin{bmatrix} t_1 & -a_1 \end{bmatrix}_J$ are units. Hence the $Z_{n,m,I,J}$ are open representable subfunctors of $Z_{n,m}$.

By [GW20, Theorem 8.9] it suffices to show that $\coprod_{I,J} Z_{n,m,I,J} \to Z_{n,m}$ is surjective, and by [Sta26, Tag 06D4] surjectivity may be checked on fields.

Let k be a field and let $(t_0, t_1 : a_0, a_1) \in Z_{n,m}(k)$ with $t_0 \in M_n(k)$ and $t_1 \in M_m(k)$. By Lemma 2.4 the matrix $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix} : k^n \to k^{n+m}$ is split injective and $\begin{bmatrix} t_1 & -a_1 \end{bmatrix} : k^{n+m} \to k^m$ is surjective, so $\operatorname{rank} \begin{bmatrix} a_0 \\ t_0 \end{bmatrix} = n$ and $\operatorname{rank} \begin{bmatrix} t_1 & -a_1 \end{bmatrix} = m$. Each therefore has an invertible maximal minor, which shows that $\coprod_{I,J} Z_{n,m,I,J}(k) \to Z_{n,m}(k)$ is surjective. □

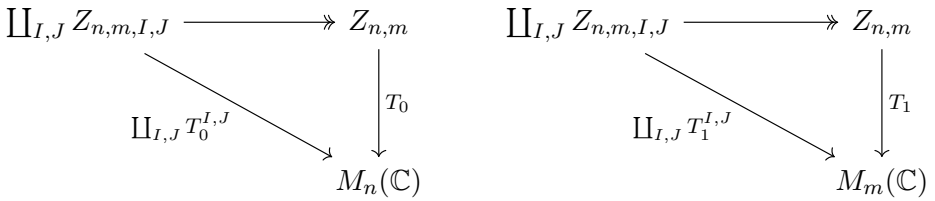
Fix $m \le n$. For $I, J \subset \{1, \dots, n+m\}$ with $|I| = n$ and $|J| = m$, define

$$\begin{aligned} T_0^{I,J}\colon Z_{n,m,I,J} &\longrightarrow M_n(\mathbb{C}), & (t_0, t_1, a_0, a_1, r_1, r_2) &\longmapsto t_0, \\ T_1^{I,J}\colon Z_{n,m,I,J} &\longrightarrow M_m(\mathbb{C}), & (t_0, t_1, a_0, a_1, r_1, r_2) &\longmapsto t_1. \end{aligned}$$

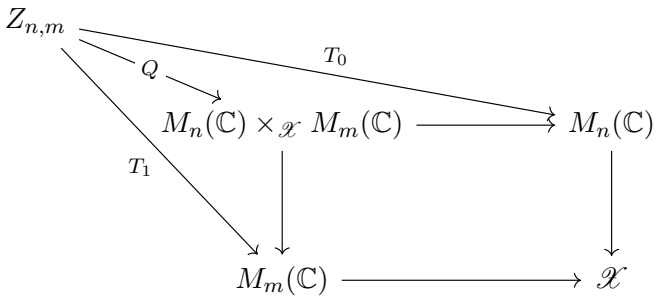
Since $\coprod_{I,J} Z_{n,m,I,J} \to Z_{n,m}$ is surjective and

$$\begin{aligned} T_0^{I,J}\big|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}} &= T_0^{I',J'}\big|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}}, \\ T_1^{I,J}\big|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}} &= T_1^{I',J'}\big|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}}, \end{aligned}$$

we obtain maps $T_0\colon Z_{n,m} \to M_n(\mathbb{C})$ and $T_1\colon Z_{n,m} \to M_m(\mathbb{C})$ making both triangles commute:



Moreover, by the universal property of the fibre product there is a unique map Q making the triangles below commute:



Lemma 2.6. *The map T_0 is smooth and surjective.*

Proof. Surjectivity of T_0 is clear from the construction of $Z_{n,m}$. Since $Z_{n,m}$ is covered by the $Z_{n,m,I,J}$, it suffices to show that $T_0^{I,J}\colon Z_{n,m,I,J} \to M_n(\mathbb{C})$ is smooth. Write

$$\begin{aligned} t_0 &= \{t_{ij}\}_{1\le i,j\le n}, & t_1 &= \{t'_{ij}\}_{1\le i,j\le m},\\ a_0 &= \{a_{ij}\}_{1\le i\le m,\ 1\le j\le n}, & a_1 &= \{a'_{ij}\}_{1\le i\le m,\ 1\le j\le n},\\ d_1 &= \det \begin{bmatrix} a_0 \\ t_0 \end{bmatrix}_I, & d_2 &= \det \begin{bmatrix} t_1 & -a_1 \end{bmatrix}_J. \end{aligned}$$

The defining equations are

$$\begin{aligned} e_{ij} &:= \sum_{k=1}^{m} t'_{ik}a_{kj} - \sum_{k=1}^{n} a'_{ik}t_{kj} = 0 \qquad \text{for } 1\le i\le m,\ 1\le j\le n,\\ z_1 &:= r_1 d_1 - 1 = 0,\\ z_2 &:= r_2 d_2 - 1 = 0, \end{aligned}$$

and

$$\frac{\partial e_{i'j'}}{\partial t'_{ij}} = \delta_{ii'}a_{jj'}, \qquad \frac{\partial e_{i'j'}}{\partial a'_{ij}} = -\delta_{ii'}t_{jj'}.$$

The Jacobian matrix J of these equations relative to $M_n(\mathbb{C})$ (i.e. with t_0 held fixed) is the $(m^2+2mn+2)\times(mn+2)$ matrix

$$J = \begin{array}{c} \\ \left(\frac{\partial}{\partial a_{ij}}\right)_{1\le i\le m,\,1\le j\le n} \\ \left(\frac{\partial}{\partial t'_{1j}}\right)_{1\le j\le m} \\ \vdots \\ \left(\frac{\partial}{\partial t'_{mj}}\right)_{1\le j\le m} \\ \left(\frac{\partial}{\partial a'_{1j}}\right)_{1\le j\le n} \\ \vdots \\ \left(\frac{\partial}{\partial a'_{mj}}\right)_{1\le j\le n} \\ \frac{\partial}{\partial r_1} \\ \frac{\partial}{\partial r_2} \end{array} \begin{array}{c} \begin{array}{ccccc} (e_{1j})_{1\le j\le n} & \cdots & (e_{mj})_{1\le j\le n} & z_1 & z_2 \end{array} \\ \begin{pmatrix} * & * & * & * & * \\ a_0 & 0 & 0 & * & * \\ 0 & \ddots & 0 & * & * \\ 0 & 0 & a_0 & * & * \\ -t_0 & 0 & 0 & * & * \\ 0 & \ddots & 0 & * & * \\ 0 & 0 & -t_0 & * & * \\ 0 & 0 & 0 & d_1 & 0 \\ 0 & 0 & 0 & 0 & d_2 \end{pmatrix} \end{array}$$

Since d_1 and d_2 are invertible and $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$ has rank n, the matrix J has rank $mn+2$, i.e. full rank, at every point of $Z_{n,m,I,J}$. So the $mn+2$ equations e_{ij}, z_1, z_2 cut out a smooth complete intersection of codimension $mn+2$ in each fibre of the projection to $M_n(\mathbb{C})$; the fibres therefore all have the same dimension and are smooth, and $T_0^{I,J}$ is flat with smooth fibres, hence smooth. □

Remark 2.7. The same computation with the roles of the two complexes exchanged – differentiating the e_{ij} with respect to the entries of t_0 and a_0 instead of t_1 and a_1 – shows that $T_1\colon Z_{n,m} \to M_m(\mathbb{C})$ is smooth and surjective as well.

Consider the additive group scheme $M_{m,n} = \operatorname{Spec}\mathbb{C}[(x_{ij})_{1\le i\le m,\,1\le j\le n}]$. For any commutative $\mathbb{C}$-algebra A there is an action of $M_{m,n}(A)$ on $Z_{n,m}(A)$ given by

$$z\cdot(t_0,t_1:a_0,a_1) = (t_0,t_1:a_0+zt_0,\ a_1+t_1z)$$

for $z\in M_{m,n}(A)$ and $(t_0,t_1:a_0,a_1)\in Z_{n,m}(A)$. This lands back in $Z_{n,m}(A)$: the first condition of Lemma 2.4 is preserved because

$$t_1(a_0+zt_0)-(a_1+t_1z)t_0 = t_1a_0-a_1t_0,$$

and the other two because $\left[\begin{smallmatrix} a_0+zt_0 \\ t_0 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} t_1 & -(a_1+t_1z) \end{smallmatrix}\right]$ differ from $\begin{bmatrix} a_0 \\ t_0 \end{bmatrix}$ and $\begin{bmatrix} t_1 & -a_1 \end{bmatrix}$ by multiplication by the invertible matrices $\left[\begin{smallmatrix} I_m & z \\ 0 & I_n \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} I_m & 0 \\ -z & I_n \end{smallmatrix}\right]$ respectively.

Proposition 2.8. *For any $N\ge 0$, every principal $\mathbb{G}_a^N$-bundle over an affine scheme is trivial. In particular this applies to $M_{m,n}\cong\mathbb{G}_a^{mn}$.*

Proof. Isomorphism classes of principal $\mathbb{G}_a^N$-bundles over a scheme X are classified by $H^1_{\text{fppf}}(X, \mathbb{G}_a^N)$. Since $\mathbb{G}_a^N$ is a smooth group scheme, its fppf and étale cohomology agree with its Zariski cohomology, and $\mathbb{G}_a^N$ is the group scheme attached to the quasi-coherent sheaf $\mathcal{O}_X^{\oplus N}$:

$$H^1_{\text{fppf}}(X, \mathbb{G}_a^N) \cong H^1_{\text{ét}}(X, \mathbb{G}_a^N) \cong H^1_{\text{Zar}}(X, \mathcal{O}_X)^{\oplus N}.$$

If X is affine then $H^1(X, \mathcal{O}_X) = 0$ by Serre vanishing, so the group is trivial and hence so is every such bundle. □

Lemma 2.9. *The quotient stack $[Z_{n,m}/M_{m,n}]$ is equivalent to $M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C})$.*

Proof. For a commutative $\mathbb{C}$-algebra A, the objects of $[Z_{n,m}/M_{m,n}](A)$ are the pairs

$$\big\{(p, f) \;\big|\; p\colon P \to \operatorname{Spec}(A) \text{ an } M_{m,n}\text{-bundle},\ f\colon P \to Z_{n,m}\ M_{m,n}\text{-equivariant}\big\}.$$

Since $M_{m,n}$ is an additive group, every principal $M_{m,n}$-bundle over an affine scheme is trivial by Proposition 2.8. Hence the objects of $[Z_{n,m}/M_{m,n}](A)$ are

$$\{p\colon M_{m,n} \times \operatorname{Spec}(A) \to Z_{n,m} \mid p \text{ is } M_{m,n}\text{-equivariant}\}.$$

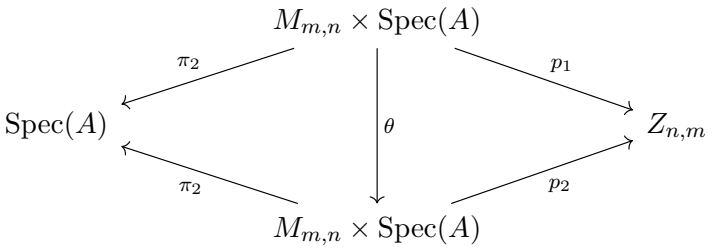
By equivariance, such a p is determined by its restriction

$$p\big|_{\{0\}\times\operatorname{Spec}(A)}\colon \{0\} \times \operatorname{Spec}(A) = \operatorname{Spec}(A) \to Z_{n,m},$$

which is an object of $Z_{n,m}(A)$. Since the objects of $Z_{n,m}(A)$ are the objects of $(M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}))(A)$, we obtain a bijection

$$\begin{aligned}[Z_{n,m}/M_{m,n}](A) &\longleftrightarrow (M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}))(A)\\ p &\longmapsto p\big|_{\{0\}\times\operatorname{Spec}(A)}.\end{aligned}$$

A morphism from p_1 to p_2 is an $M_{m,n}$-equivariant isomorphism θ making the following diagram commute:



where π_2 is the projection to the second coordinate. Since the left-hand triangle commutes we may write $\theta(z, x) = (\theta_1(z, x), x)$ for an $M_{m,n}$-equivariant map $\theta_1\colon M_{m,n} \times \operatorname{Spec}(A) \to M_{m,n}$. Such a θ_1 is determined by $\theta_1(0, -)\colon \operatorname{Spec}(A) \to M_{m,n}$, i.e. by an element of $M_{m,n}(A)$. It must satisfy

$$p_2(\theta_1(0, x), x) = p_1(0, x) \qquad \text{for all } x \in \operatorname{Spec}(A).$$

Suppose p_1 and p_2 correspond to $(t_0, t_1 : a_0, a_1)$ and $(t_0', t_1' : a_0', a_1')$ respectively, i.e.

$$p_1\big|_{\{0\}\times \mathrm{Spec}(A)} = (t_0, t_1, a_0, a_1),$$
$$p_2\big|_{\{0\}\times \mathrm{Spec}(A)} = (t_0', t_1', a_0', a_1').$$

Then

$$\begin{aligned}\mathrm{Mor}(p_1, p_2) &= \{z \in M_{m,n}(A) \mid p_2(z, x) = p_1(0, x)\ \forall x \in \mathrm{Spec}(A)\}\\ &= \{z \in M_{m,n}(A) \mid z \cdot p_2(0, x) = p_1(0, x)\ \forall x \in \mathrm{Spec}(A)\}\\ &= \{z \in M_{m,n}(A) \mid z \cdot (t_0', t_1', a_0', a_1') = (t_0, t_1, a_0, a_1)\}\\ &= \{z \in M_{m,n}(A) \mid (t_0', t_1', a_0' + zt_0', a_1' + t_1'z) = (t_0, t_1, a_0, a_1)\}.\end{aligned}$$

So $\mathrm{Mor}(p_1, p_2) = \emptyset$ if $(t_0, t_1) \neq (t_0', t_1')$, while if $(t_0, t_1) = (t_0', t_1')$ then

$$\mathrm{Mor}(p_1, p_2) = \{z \in M_{m,n}(A) \mid a_0 - a_0' = zt_0 \text{ and } a_1 - a_1' = t_1 z\},$$

which is exactly the set of chain homotopies above. Hence $[Z_{n,m}/M_{m,n}](A)$ and $(M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}))(A)$ are equivalent categories for every A, which proves the lemma. □

Corollary 2.10. *The map $M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}) \to M_m(\mathbb{C})$ is smooth and surjective.*

Proof. The map $Z_{n,m} \to [Z_{n,m}/M_{m,n}] \cong M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C})$ is smooth and surjective, and $T_1 \colon Z_{n,m} \to M_m(\mathbb{C})$ is smooth and surjective by Remark 2.7; smoothness descends along smooth surjections, so $M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}) \to M_m(\mathbb{C})$ is smooth and surjective. □

Together with Lemma 2.3, this shows that the maps $f_n \colon M_n(\mathbb{C}) \to \mathscr{X}$ form a covering.

3. Understanding $Z_{n,m}$

Let us work with the case $A = \mathbb{C}$. If t_0 and t_1 have different corank then the associated complexes are not quasi-isomorphic. Take

$$t_0 = \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix}, \qquad t_1 = \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix},$$
$$a_0 = \{x_{ij}\}_{1\le i\le m,\, 1\le j\le n}, \qquad a_1 = \{y_{ij}\}_{1\le i\le m,\, 1\le j\le n}.$$

Then all quasi-isomorphisms (a_0, a_1) between t_0 and t_1 are found from

$$\begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix} \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix} = \begin{bmatrix} y_{11} & \cdots & y_{1n} \\ \vdots & \ddots & \vdots \\ y_{m1} & \cdots & y_{mn} \end{bmatrix} \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix},$$

together with the requirement that

$$a_0|_{\ker(t_0)} \colon \ker(t_0) \to \ker(t_1),$$
$$[a_1] \colon \mathbb{C}^n/\operatorname{im}(t_0) \to \mathbb{C}^m/\operatorname{im}(t_1)$$

be isomorphisms. Solving for x_{ij} and y_{ij} shows that

$$a_0 = \begin{bmatrix} X_k & X \\ 0 & Z \end{bmatrix} \qquad \text{and} \qquad a_1 = \begin{bmatrix} Y_k & 0 \\ Y & Z \end{bmatrix},$$

where $X_k, Y_k \in \mathrm{GL}_k(\mathbb{C})$, $X \in M_{k,n-k}(\mathbb{C})$, $Y \in M_{m-k,k}(\mathbb{C})$ and $Z \in M_{m-k,n-k}(\mathbb{C})$.

Since over a field every matrix can be row- and column-reduced to one of the form $\begin{bmatrix} 0_k & 0 \\ 0 & I_{k'} \end{bmatrix}$, we define

$$Z_{n,m}(\mathbb{C}) := \coprod_{k=0}^{\min\{n,m\}} Z_{n,m,k},$$

where

$$\begin{aligned} Z_{n,m,k} := \Big\{ \Big(& Q \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix} P^{-1},\ R \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix} S^{-1}, \\ & S \begin{bmatrix} X_k & X \\ 0 & Z \end{bmatrix} P^{-1},\ R \begin{bmatrix} Y_k & 0 \\ Y & Z \end{bmatrix} Q^{-1} \Big) \ \Big| \\ & P, Q \in \mathrm{GL}_n(\mathbb{C}),\ R, S \in \mathrm{GL}_m(\mathbb{C}),\ X_k, Y_k \in \mathrm{GL}_k(\mathbb{C}), \\ & Z \in M_{m-k,n-k}(\mathbb{C}),\ X \in M_{k,n-k}(\mathbb{C}),\ Y \in M_{m-k,k}(\mathbb{C}) \Big\}. \end{aligned}$$

Let

$$\mathcal{G}_{n,m} := \big(\mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_m(\mathbb{C}) \times \mathrm{GL}_m(\mathbb{C})\big) \ltimes M_{m,n}(\mathbb{C}),$$

with group law

$$(P', Q', R', S', x') \cdot (P, Q, R, S, x) = (P'P,\ Q'Q,\ R'R,\ S'S,\ x + S^{-1}x'Q).$$

Lemma 3.1. *$\mathcal{G}_{n,m}$ acts on $Z_{n,m}(\mathbb{C})$ by*

$$\begin{aligned} &(P, Q, R, S, x) \cdot (t_0, t_1, a_0, a_1) \\ &\qquad = \big(Qt_0P^{-1},\ Rt_1S^{-1},\ S(a_0 + xt_0)P^{-1},\ R(a_1 + t_1x)Q^{-1}\big). \end{aligned}$$

Proof. For $g_1 = (P, Q, R, S, x)$, $g_2 = (P', Q', R', S', x') \in \mathcal{G}_{n,m}$ and $z = (t_0, t_1, a_0, a_1) \in Z_{n,m}(\mathbb{C})$,

$$\begin{aligned} g_2 \cdot (g_1 \cdot z) &= (P', Q', R', S', x') \cdot \big(Qt_0P^{-1},\ Rt_1S^{-1}, \\ &\qquad S(a_0 + xt_0)P^{-1},\ R(a_1 + t_1x)Q^{-1}\big) \\ &= \big(Q'Qt_0P^{-1}(P')^{-1},\ R'Rt_1S^{-1}(S')^{-1}, \\ &\qquad S'\big(S(a_0 + xt_0)P^{-1} + x'Qt_0P^{-1}\big)(P')^{-1}, \\ &\qquad R'\big(R(a_1 + t_1x)Q^{-1} + Rt_1S^{-1}x'\big)(Q')^{-1}\big) \\ &= \big((Q'Q)t_0(P'P)^{-1},\ (R'R)t_1(S'S)^{-1}, \\ &\qquad (S'S)\big(a_0 + (x + S^{-1}x'Q)t_0\big)(P'P)^{-1}, \\ &\qquad (R'R)\big(a_1 + t_1(x + S^{-1}x'Q)\big)(Q'Q)^{-1}\big) \\ &= (P'P, Q'Q, R'R, S'S,\ x + S^{-1}x'Q) \cdot z \\ &= (g_2 \cdot g_1) \cdot z. \end{aligned}$$

□

Lemma 3.2. *The $\mathcal{G}_{n,m}$-orbits in $Z_{n,m}(\mathbb{C})$ are precisely the $Z_{n,m,k}$, $0 \le k \le \min\{n, m\}$.*

Proof. Let

$$z = \Big(Q \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix} P^{-1},\ R \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix} S^{-1}, \\ S \begin{bmatrix} X_k & X \\ 0 & Z \end{bmatrix} P^{-1},\ R \begin{bmatrix} Y_k & 0 \\ Y & Z \end{bmatrix} Q^{-1}\Big)$$

be an element of $Z_{n,m,k}$, for some $P, Q \in \mathrm{GL}_n(\mathbb{C})$, $R, S \in \mathrm{GL}_m(\mathbb{C})$, $X_k, Y_k \in \mathrm{GL}_k(\mathbb{C})$, $Z \in M_{m-k,n-k}(\mathbb{C})$, $X \in M_{k,n-k}(\mathbb{C})$ and $Y \in M_{m-k,k}(\mathbb{C})$. Set

$$g \quad = \quad \left(P \begin{bmatrix} X_k^{-1} & 0 \\ 0 & I_{n-k} \end{bmatrix}, \quad Q, \quad R \begin{bmatrix} Y_k & 0 \\ 0 & I_{m-k} \end{bmatrix}, \quad S, \begin{bmatrix} 0 & X \\ Y & Z \end{bmatrix}\right)$$

and

$$z_0 = \left(\begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix}, \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix}, \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}\right).$$

Then $g \cdot z_0 = z$, so $Z_{n,m,k}$ is contained in a single $\mathcal{G}_{n,m}$-orbit. Since the action does not change the corank of t_0 or of t_1, distinct k give distinct orbits, which proves the lemma. □

For $n, m \in \mathbb{N}$, let $\pi_1^{n,m} \colon M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}) \to M_n(\mathbb{C})$ and $\pi_2^{n,m} \colon M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}) \to M_m(\mathbb{C})$ be the projections, so that the following diagram commutes:

$$\begin{array}{ccc} M_n(\mathbb{C}) \times_{\mathscr{X}} M_m(\mathbb{C}) & \xrightarrow{\pi_1^{n,m}} & M_n(\mathbb{C}) \\ \downarrow{\scriptstyle \pi_2^{n,m}} & & \downarrow{\scriptstyle f_n} \\ M_m(\mathbb{C}) & \xrightarrow{f_m} & \mathscr{X} \end{array}$$

Lemma 3.3. *For $n, m \in \mathbb{N}$ and $i = 1, 2$, the fibres of $\pi_i^{n,m}$ are connected.*

Corollary 3.4. *For $n, m \in \mathbb{N}$ with $n \leq m$, the functor $(\pi_1^{n,m})^*$ is fully faithful.*

4. Perverse sheaves

Let $\mathscr{X}_n := f_n(M_n(\mathbb{C}))$ be the open substack of $\mathscr{X}$ that is the image of f_n. Since $f_n \colon M_n(\mathbb{C}) \to \mathscr{X}_n$ is smooth and surjective, perverse sheaves on $\mathscr{X}_n$ may be defined by descent.

Notation 4.1. Recall from the introduction that $G_n = \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C})$ acts on $M_n(\mathbb{C})$ by $(P, Q) \cdot t = QtP^{-1}$. By Gaussian elimination the orbits of this action are the locally closed subvarieties

$$\mathcal{O}_k = \{\, t \in M_n(\mathbb{C}) : \mathrm{corank}(t) = k \,\}, \qquad 0 \leq k \leq n,$$

with $\overline{\mathcal{O}_k} = \bigcup_{l \geq k} \mathcal{O}_l$. We write $\mathrm{Perv}_{G_n}(M_n(\mathbb{C}))$ for the category of G_n-equivariant perverse sheaves on $M_n(\mathbb{C})$ with $\mathbb{C}$-coefficients and middle perversity, constructible with respect to this stratification, and we set

$$IC_k := IC(\overline{\mathcal{O}_k}, \mathbb{C}), \qquad 0 \leq k \leq n,$$

for its simple objects.

Definition 4.2. A perverse sheaf on $\mathscr{X}_n$ is a pair (F, ϕ), where $F \in \mathrm{Perv}_{G_n}(M_n(\mathbb{C}))$ and ϕ is an isomorphism

$$(\pi_1^{n,n})^* F \overset{\phi}{\cong} (\pi_2^{n,n})^* F$$

satisfying the descent datum of [Ach21, Definition 3.7.1]. These form a category $\mathrm{Perv}(\mathscr{X}_n)$.

Since the fibres of f_n are connected, the functor

$$\mathrm{Perv}_{G_n}(M_n(\mathbb{C})) \hookrightarrow \mathrm{Perv}(\mathscr{X}_n)$$

is fully faithful.

Theorem 4.3. *For $n \in \mathbb{N}$,*

$$\mathrm{Perv}(\mathscr{X}_n) \cong \mathrm{Perv}_{G_n}(M_n(\mathbb{C})).$$

Proof. Since the fibres of f_n are connected, we get a fully faithful embedding

$$\mathrm{Perv}_{G_n}(M_n(\mathbb{C})) \hookrightarrow \mathrm{Perv}(\mathscr{X}_n).$$

It therefore suffices to show that the projective covers of the IC_k of Notation 4.1 lie in $\mathrm{Perv}(\mathscr{X}_n)$.

For a projective cover P there is an isomorphism $(\pi_1^{n,n})^*P \cong (\pi_2^{n,n})^*P$. Since the isomorphism is a scalar, the cocycle condition holds up to a scalar, and we may normalise to make it hold exactly.

For an arbitrary perverse sheaf, the diagram

$$\begin{array}{ccccccc}
F & \xrightarrow{(\pi_2^{n,n})^*} & (\pi_2^{n,n})^*F & \xrightarrow{\sim} & (\pi_1^{n,n})^*F & \xleftarrow{(\pi_1^{n,n})^*} & F \\
\uparrow & & \uparrow{\scriptstyle \mathrm{coker}} & & \uparrow{\scriptstyle \mathrm{coker}} & & \uparrow \\
P^\bullet & \xrightarrow[(\pi_2^{n,n})^*]{} & (\pi_2^{n,n})^*P^\bullet & \xrightarrow[\sim]{} & (\pi_1^{n,n})^*P^\bullet & \xleftarrow[(\pi_1^{n,n})^*]{} & P^\bullet
\end{array}$$

produces the required isomorphism. Hence the embedding is also essentially surjective, so it is an equivalence of categories. □

Definition 4.4. $\mathrm{Perv}(\mathscr{X}) := \varprojlim_i \mathrm{Perv}(\mathscr{X}_i)$.

Theorem 4.5. $\mathrm{Perv}(\mathscr{X}) \cong \mathrm{Rep}(AA_\infty)$.

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